Valence Alignment in Human-AI Story Co-Creation

Analysis of Field Data and AI-AI Baseline Comparison

PENPAL Project

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Contents

0.1 Summary

- Emotional alignment is asymmetric: the model reliably aligns to the user (User → AI), while humans do not mirror the preceding AI tone.
- Field data shows greater variance in valence and weaker overall coordination compared to the simulated baseline.
- Overall, affective dynamics in the field are primarily model-driven rather than mutually convergent.

1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

```
## Field data dimensions: 870 17
```

```
## Simulated data dimensions: 30 12
```

```
##
## Field data columns:
```

```
##  [1] "timestamp"                  "user"
##  [3] "ai"                         "conversation_id"
##  [5] "respondent_id"              "interaction_count"
##  [7] "llm_type"                   "edit_distance"
##  [9] "turn"                       "starter"
## [11] "user_embedding"             "ai_embedding"
## [13] "pct_turn"                   "user_sentiment_score"
## [15] "ai_sentiment_score"         "user_sentiment_projection"
## [17] "ai_sentiment_projection"
```

```
##
## Simulated data columns:
```

```
##  [1] "turn"                 "user"                 "ai"
##  [4] "client_id"            "workshop_id"          "conversation_id"
##  [7] "timestamp"            "user_embedding"       "ai_embedding"
## [10] "pct_turn"             "user_sentiment_score" "ai_sentiment_score"
```

1.2 1.2 Data Preprocessing

```
##
## === Field Data Summary ===
```

```
## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1              87     696                    8           -0.0891         0.00381
```

```
##
## === Simulated Data Summary ===
```

```
## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1               6      30                    5            -0.407          -0.296
```

1.3 1.3 Compute Lagged Variables and Cosine

```
## === Lag Variable Verification ===

## Field data - Turn 1 with non-NA lag (should be 0): 0

## Simulated data - Turn 1 with non-NA lag (should be 0): 0
```

2 2. Descriptive Statistics

2.1 2.1 Token Count Statistics

```
## === Field Data Token Statistics ===

## # A tibble: 1 x 5
##   mean_user_tokens sd_user_tokens mean_ai_tokens sd_ai_tokens mean_token_ratio
##              <dbl>          <dbl>          <dbl>        <dbl>            <dbl>
## 1             25.4           5.28           29.1         7.45            0.913
```

2.2 2.2 Valence Distribution Statistics

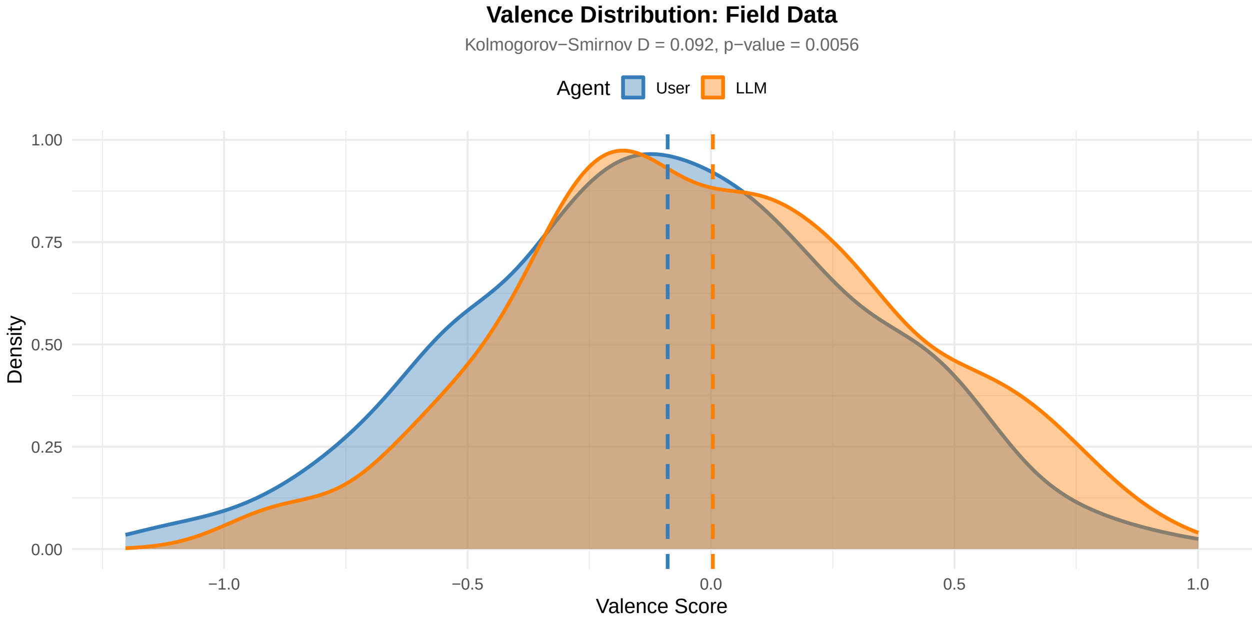
```
## # A tibble: 4 x 5
##   data_source agent mean_valence sd_valence median_valence
##   <chr>       <chr>        <dbl>      <dbl>          <dbl>
## 1 Field       ai         0.00381      0.397        -0.0212
## 2 Field       user      -0.0891       0.403        -0.0895
## 3 Simulated   ai        -0.296        0.681        -0.487
## 4 Simulated   user      -0.407        0.529        -0.504
```

3 3. Valence Distribution Analysis

3.1 3.1 Valence Distribution Difference Test

```
## === Valence Difference Model Coefficients ===

##             Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)  -0.0891     0.0269  108.6627 -3.3080   0.0013
## agentLLM      0.0929     0.0179 1304.0000  5.1811   0.0000
```

3.2 3.2 Combined Valence Distribution Plot




```
##
## === Kolmogorov-Smirnov Test: User vs LLM Valence Distribution ===
##
##  Asymptotic two-sample Kolmogorov-Smirnov test
##
## data:  df_field_long %>% filter(agent == "User") %>% pull(valence) and df_field_long %>% filter(agen
## D = 0.091954, p-value = 0.005561
## alternative hypothesis: two-sided
```

4 4. Valence Alignment Analysis

4.1 4.1 Per-Story Correlation (Same Turn)

```
## === Same-Turn Alignment Summary ===
## # A tibble: 2 x 5
##   data_source mean_rho sd_rho median_rho n_stories
##   <chr>          <dbl>  <dbl>      <dbl>     <int>
## 1 Field          0.232  0.389      0.290        87
## 2 Simulated     -0.246  0.440     -0.314         6
```

4.2 4.2 Per-Story Correlation (Previous Turn)

```
## === Previous-Turn Alignment Summary ===
## # A tibble: 2 x 5
##   data_source mean_rho sd_rho median_rho n_stories
##   <chr>          <dbl>  <dbl>      <dbl>     <int>
## 1 Field         0.0908  0.395      0.123        87
## 2 Simulated     0.533   0.224      0.546         6
```

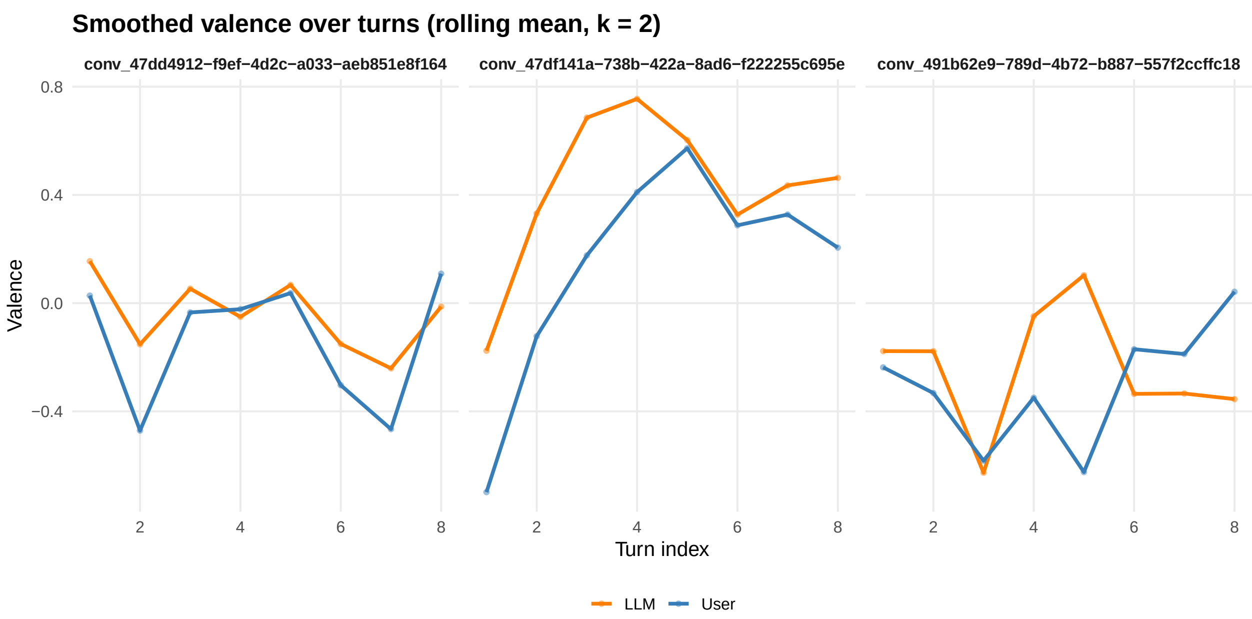
4.3 4.3.1 All turns story correlation (same turn)

```
## === Same-Turn All Turns Alignment Summary ===

## # A tibble: 2 x 5
##   data_source mean_rho sd_rho median_rho n_stories
##   <chr>          <dbl>  <dbl>      <dbl>     <int>
## 1 Field          0.427     NA      0.427         1
## 2 Simulated     -0.110     NA     -0.110         1
```

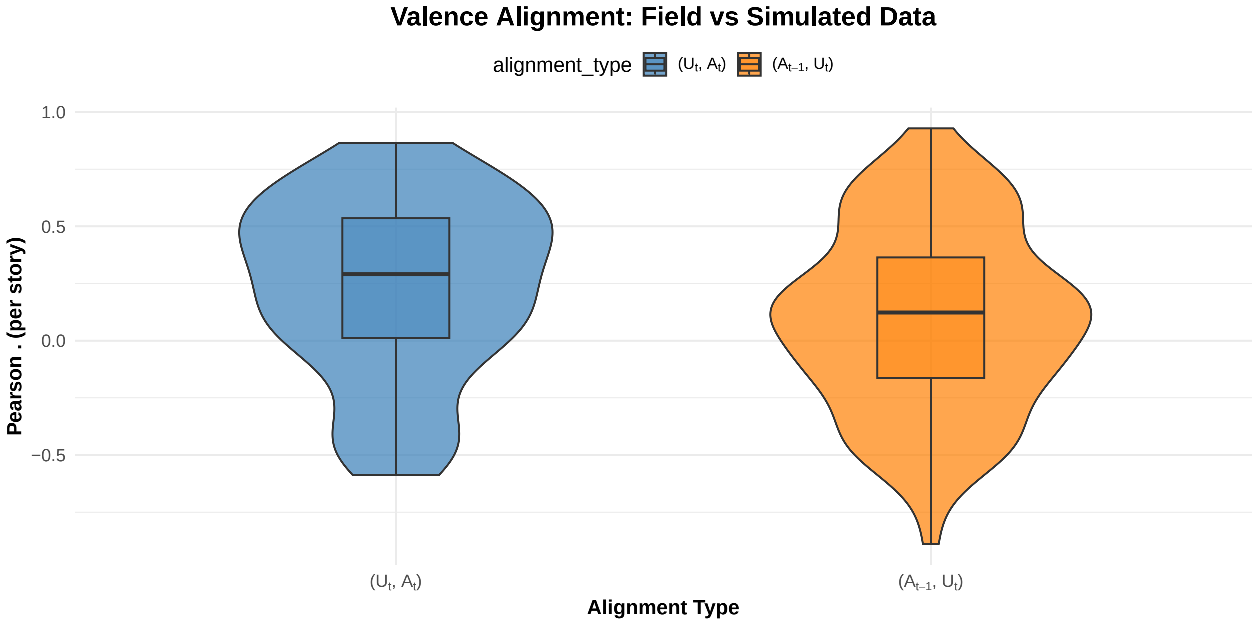
4.4 4.3.2 All turns story correlation (Previous Turn)

```
## === Previous-Turn Aggregate Correlation ===

## Field: Turn 1 observations (excluded): 0

## Field: Valid lag observations: 609

## Simulated: Turn 1 observations (excluded): 6

## Simulated: Valid lag observations: 24

## === Previous-Turn All Turns Alignment Summary ===

## # A tibble: 2 x 6
##   data_source mean_rho sd_rho median_rho n_total n_valid
##   <chr>          <dbl>  <dbl>      <dbl>   <int>   <int>
## 1 Field          0.365     NA      0.365     696     609
## 2 Simulated      0.425     NA      0.425      30      24
```

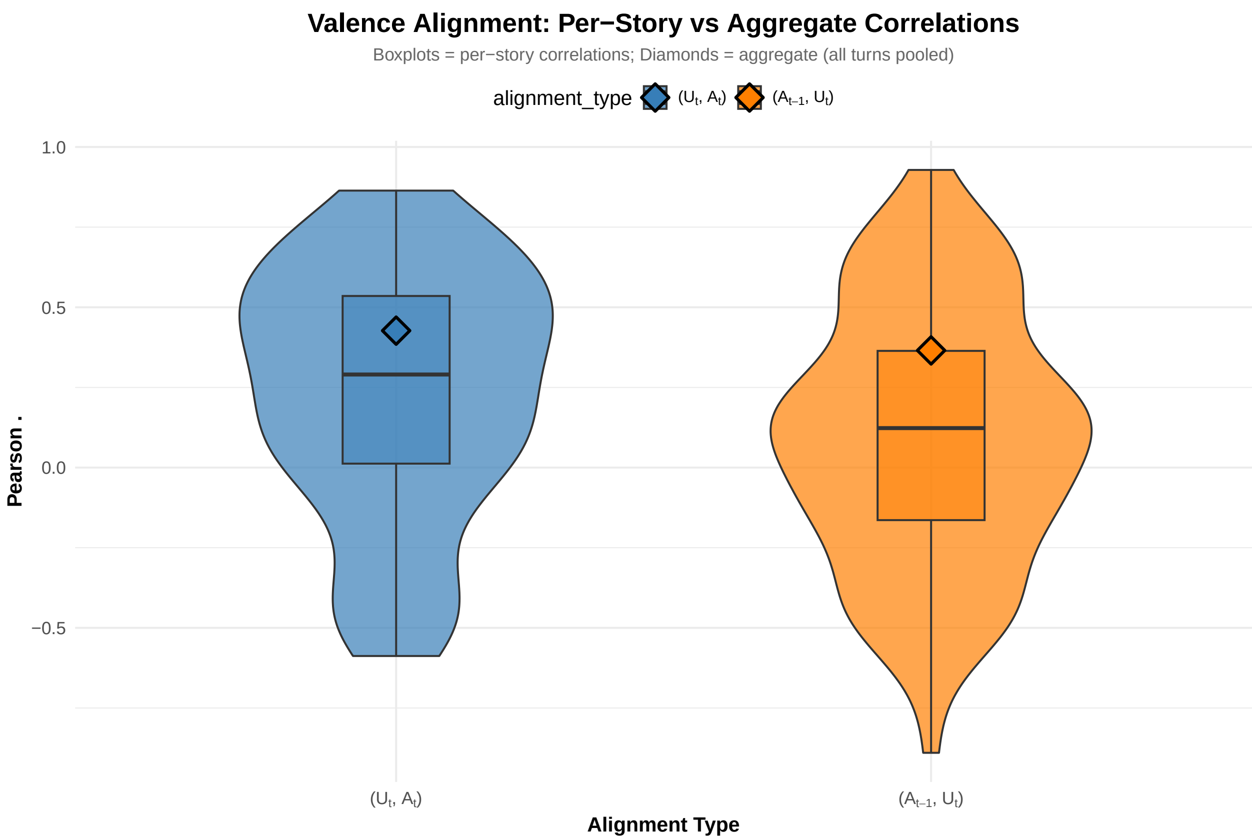
4.5 4.3.3 Probe Sentiment Narrative Development



4.6 4.4.1 Alignment Comparison Plot



4.7 4.4.2 Comparison: Per-Story vs Aggregate Correlations




```
##
## === Per-Story vs Aggregate Correlation Comparison ===
## # A tibble: 4 x 4
```

```
##   data_source alignment_type `Per-Story Mean` `Aggregate All Turns`
##   <chr>       <fct>                   <dbl>                 <dbl>
## 1 Field       same_turn               0.232                 0.427
## 2 Field       prev_turn               0.0908                0.365
## 3 Simulated   same_turn              -0.246                NA
## 4 Simulated   prev_turn               0.533                NA
```

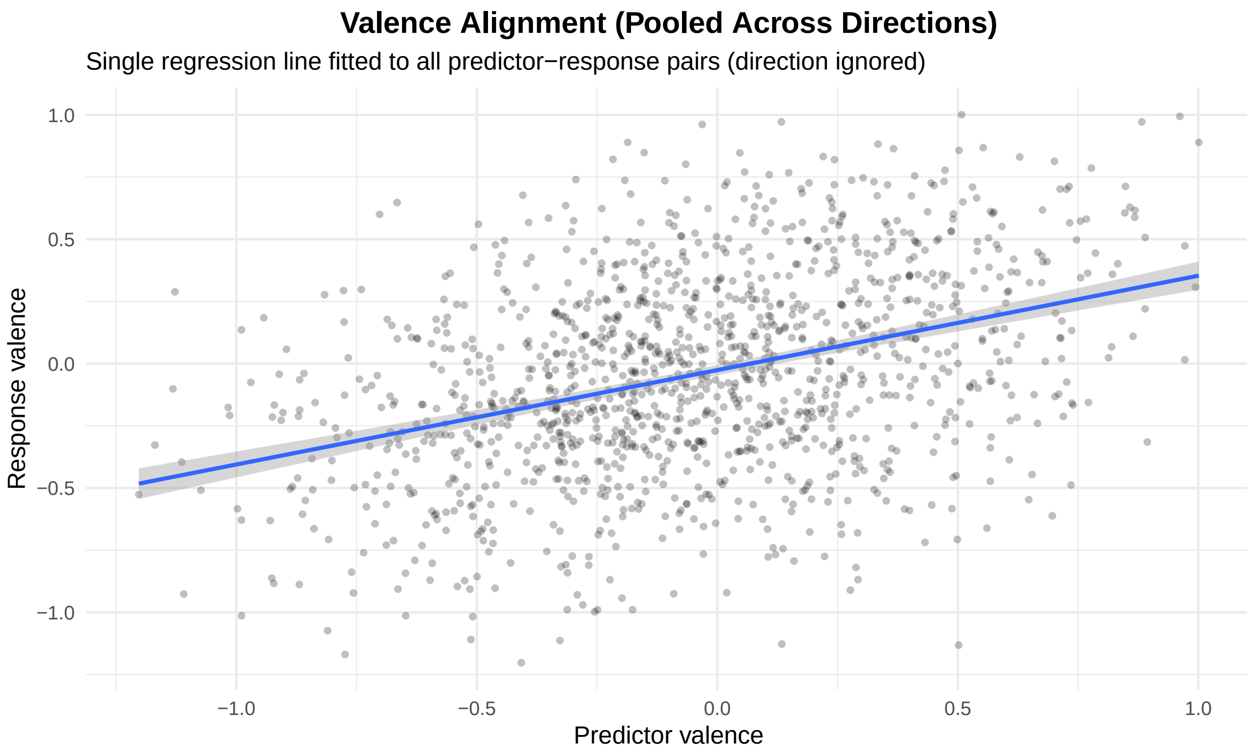
```
## === Directional Alignment Long-Format Data ===
```

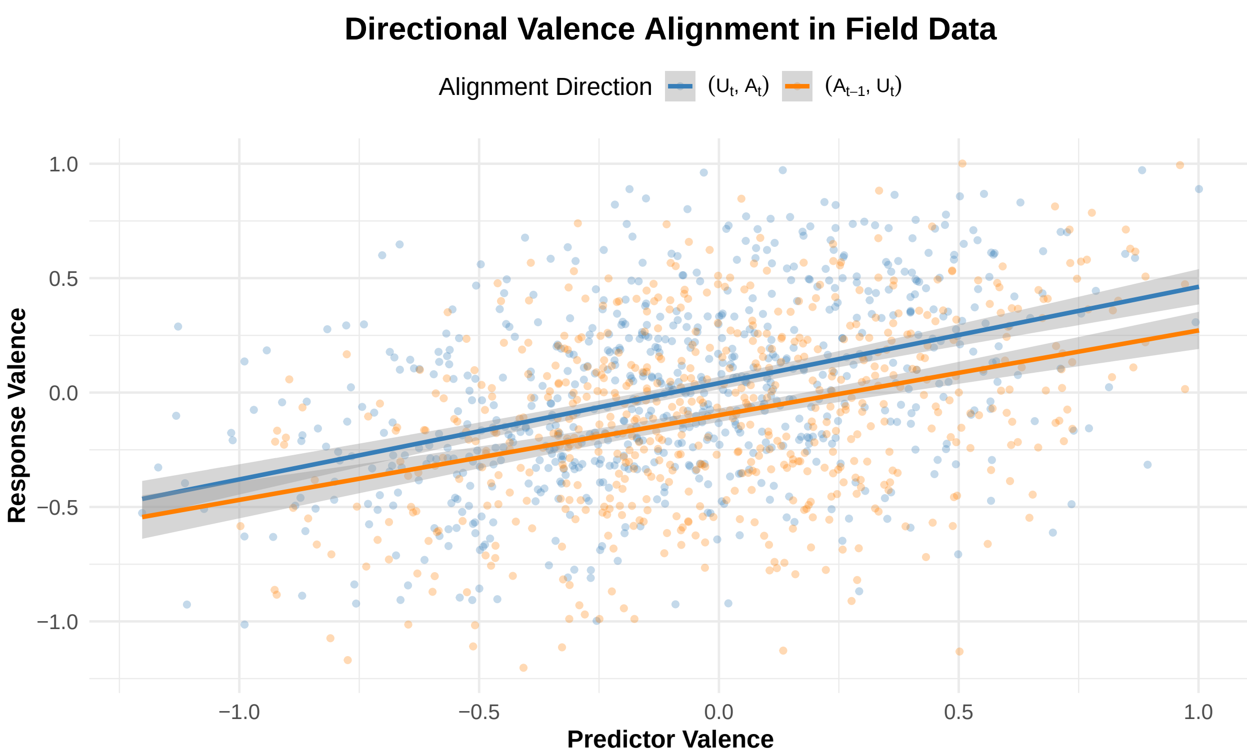
```
## # A tibble: 2 x 5
##   alignment_direction n_rows mean_predictor mean_response n_conversations
##   <fct>                <int>          <dbl>         <dbl>           <int>
## 1 User_to_AI             696       -0.0891        0.00381              87
## 2 AI_to_User             609       -0.00336      -0.100                87
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: response_z ~ predictor_z * alignment_direction + (1 + predictor_z |
##     conversation_id)
##    Data: df_std
##
## REML criterion at convergence: 3494.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.71381 -0.74858  0.01686  0.72356  2.57485
##
## Random effects:
##  Groups          Name        Variance Std.Dev. Corr
##  conversation_id (Intercept) 0.000000 0.00000
##                  predictor_z 0.005072 0.07121   NaN
##  Residual                    0.836560 0.91464
## Number of obs: 1305, groups:  conversation_id, 87
##
## Fixed effects:
##                                  Estimate Std. Error         df t value
## (Intercept)                     8.139e-18  2.538e-02  1.215e+03   0.000
## predictor_z                     1.612e-01  2.833e-02  8.693e+01   5.691
## alignment_direction1           -1.633e-17  2.538e-02  1.215e+03   0.000
## predictor_z:alignment_direction1  7.040e-02  2.728e-02  1.215e+03   2.581
##                                Pr(>|t|)
## (Intercept)                     1.00000
## predictor_z                    1.68e-07 ***
## alignment_direction1            1.00000
## predictor_z:alignment_direction1  0.00998 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) prdct_ algn_1
## predictor_z  0.000
## algnmnt_dr1 -0.067  0.000
## prdctr_z:_1  0.000 -0.074  0.000
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
##  alignment_direction predictor_z.trend     SE  df lower.CL upper.CL
##  User_to_AI                     0.2316 0.0378 264   0.1571    0.306
##  AI_to_User                     0.0908 0.0408 342   0.0106    0.171
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
##  contrast                 estimate     SE   df t.ratio p.value
##  AI_to_User - User_to_AI    -0.141 0.0546 1129  -2.581  0.0100
##
## Degrees-of-freedom method: kenward-roger
```





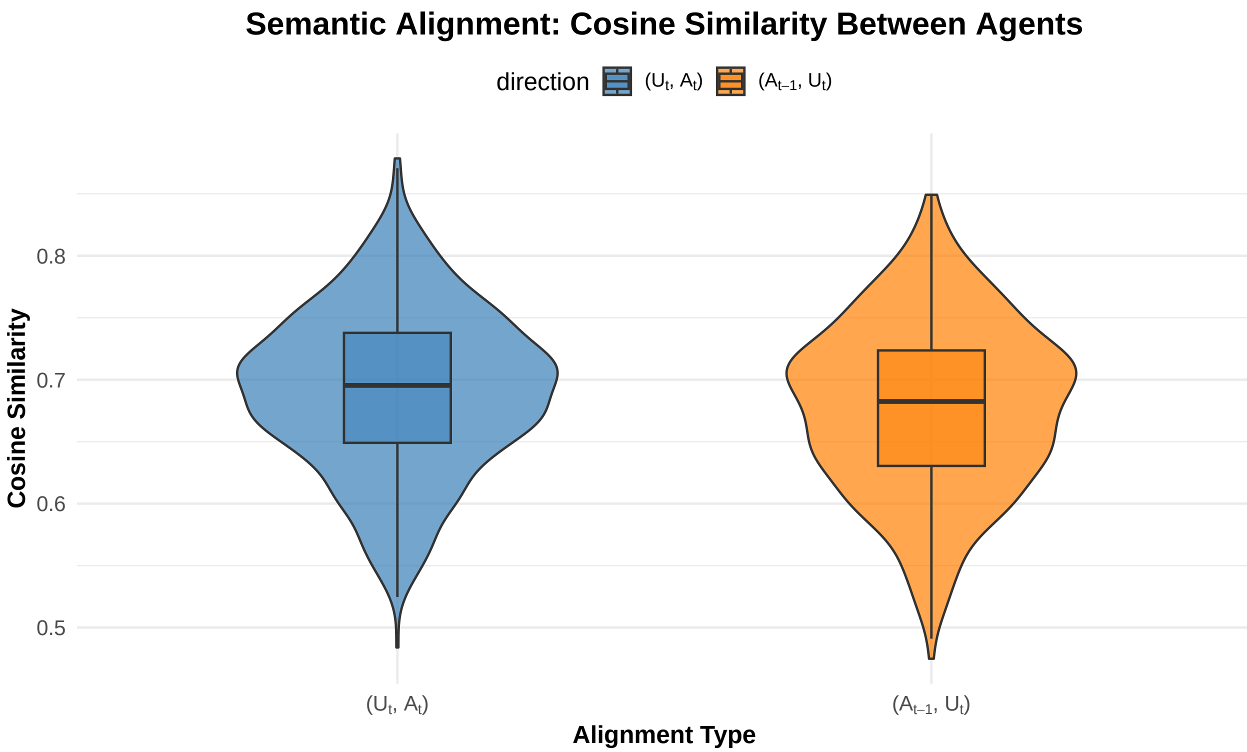
4.8 4.6 Statistical Tests: Alignment Differences

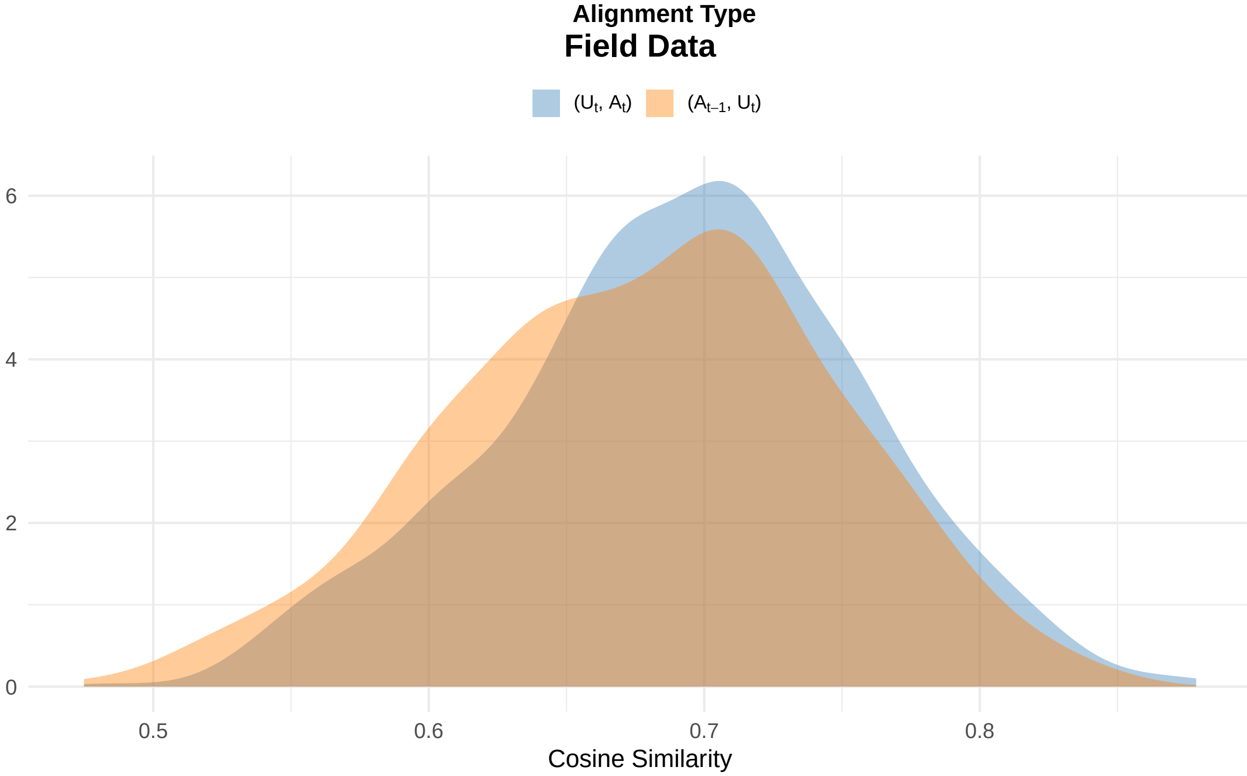
```
##
## === One-Sample Tests Against Zero ===

## # A tibble: 4 x 6
##   data_source alignment_type t_stat     p_value mean_rho     n
##   <chr>       <fct>           <dbl>       <dbl>    <dbl> <int>
## 1 Field       same_turn        5.55 0.000000312   0.232     87
## 2 Field       prev_turn        2.14 0.0349        0.0908    87
## 3 Simulated   same_turn       -1.37 0.229        -0.246      6
## 4 Simulated   prev_turn        5.84 0.00209       0.533      6

##
## === Paired T-Test: Directional Asymmetry (Same vs Prev) ===

## Field Data: Paired t-test (Fisher z-transformed Same Turn - Previous Turn)

##
##  Paired t-test
##
## data:  atanh(pairs_field$rho_same) and atanh(pairs_field$rho_prev)
## t = 2.2023, df = 86, p-value = 0.03032
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  0.01613391 0.31536192
## sample estimates:
## mean difference
##       0.1657479

##
## --- Interpretation (Back-transformed means) ---
```

```
## Mean Same-Turn r (User->AI): 0.272

## Mean Prev-Turn r (AI->User): 0.112

##
## Simulated Data: Paired t-test (Fisher z-transformed Same Turn - Previous Turn)

##
##  Paired t-test
##
## data:  atanh(pairs_sim$rho_same) and atanh(pairs_sim$rho_prev)
## t = -3.4821, df = 5, p-value = 0.01762
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  -1.6343524 -0.2461251
## sample estimates:
## mean difference
##      -0.9402387
```

4.9 Cosine Distribution Difference Test

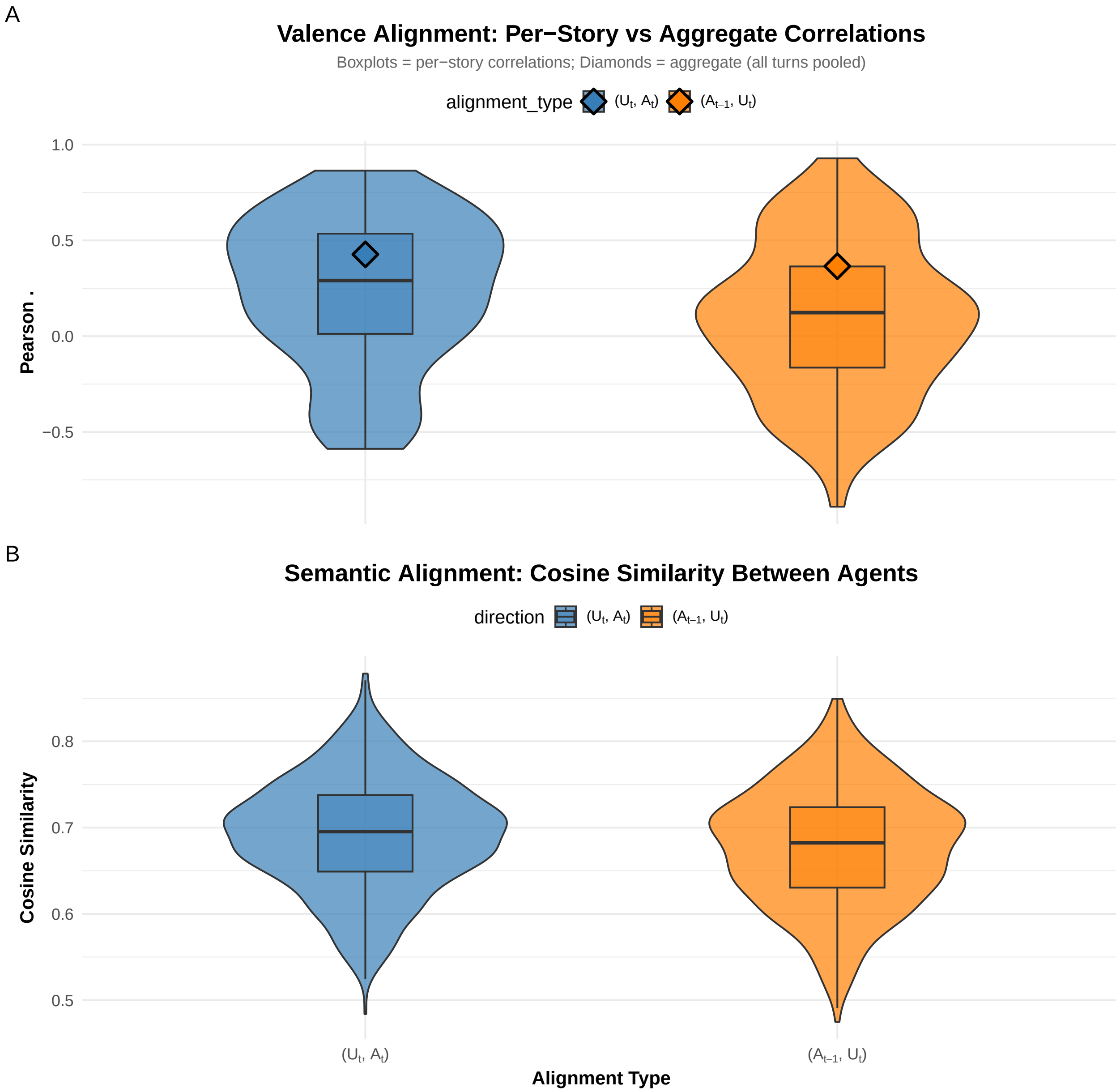
```
##
## === Cosine Sim: User (Self vs Interlocutor) ===

##                       Estimate Std. Error      df  t value Pr(>|t|)
## (Intercept)             0.6897     0.0041 86.0000 169.6669   0.0000
## directionInterlocutor  -0.0117     0.0040 85.9997  -2.9192   0.0045

##
## === Cosine Sim: AI (Self vs Interlocutor) ===

##                       Estimate Std. Error      df  t value Pr(>|t|)
## (Intercept)             0.6860     0.0041 86.0120 167.2791   0.0000
## directionInterlocutor   0.0054     0.0036 86.2026   1.5244   0.1311

##
## === Cosine Sim: General Direction Comparison ===

##                    Estimate Std. Error       df  t value Pr(>|t|)
## (Intercept)          0.6914     0.0036  94.1491 192.3986    0e+00
## directionprev_turn  -0.0134     0.0034 691.2310  -3.9250    1e-04
```

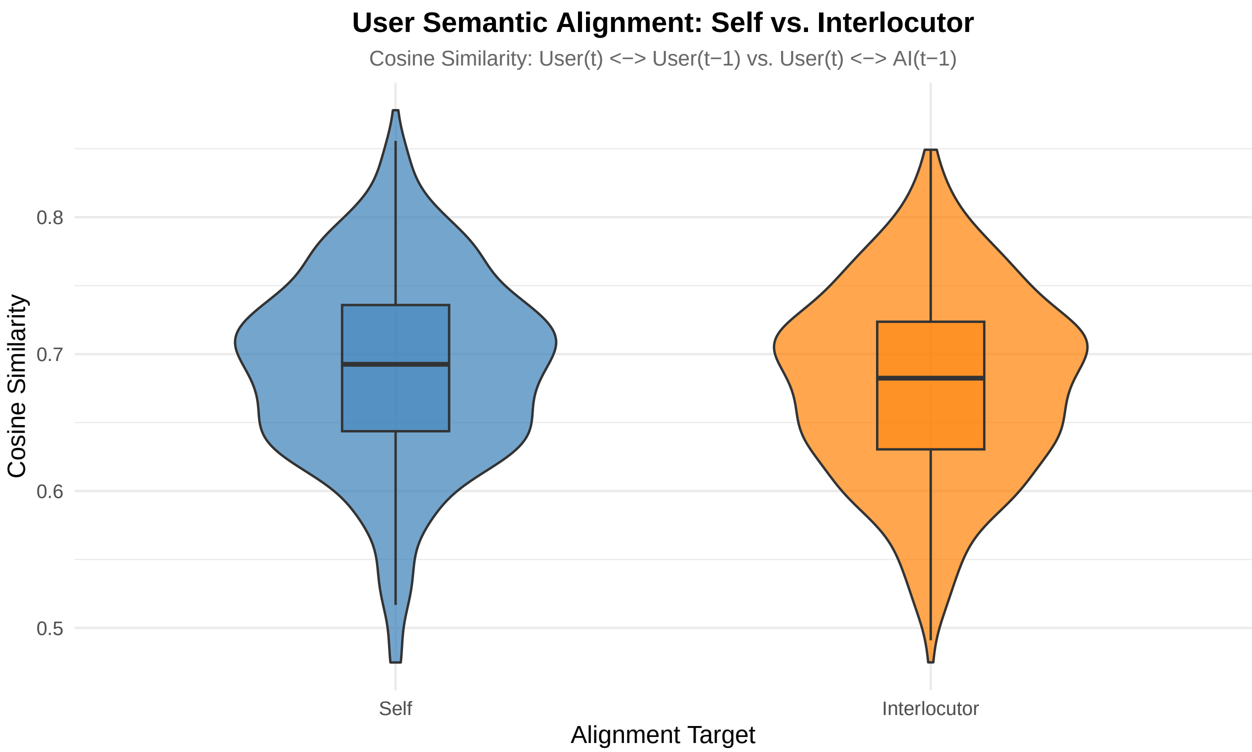




Combined Alignment Plot

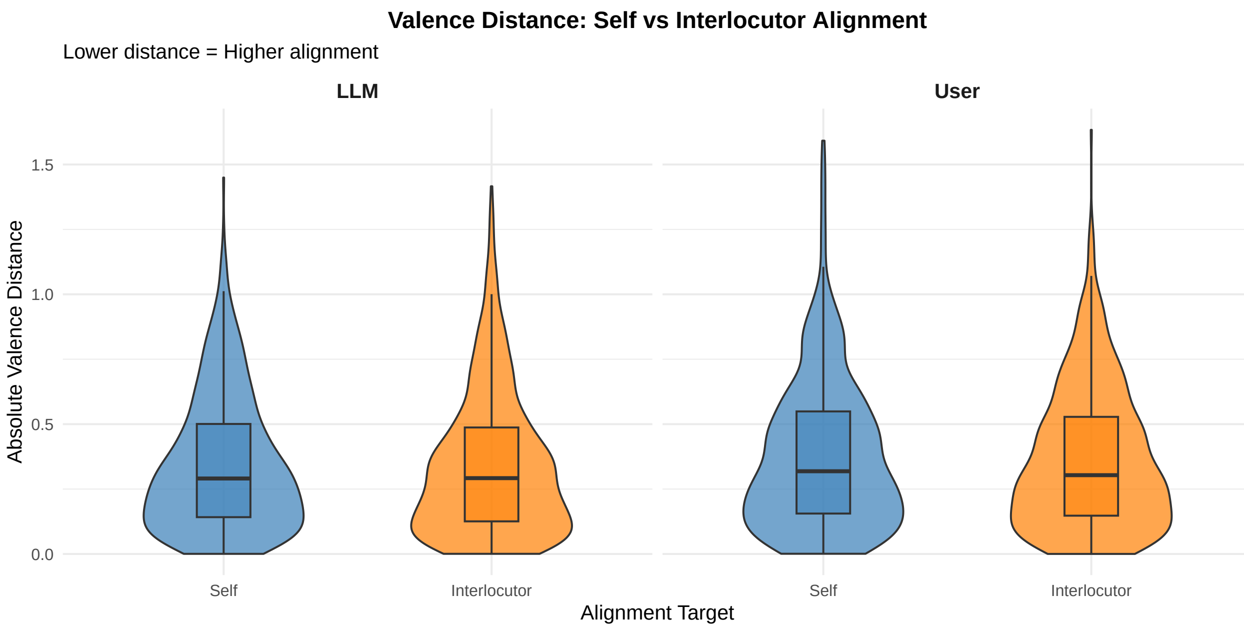
Comparison of Valence and Semantic Alignment





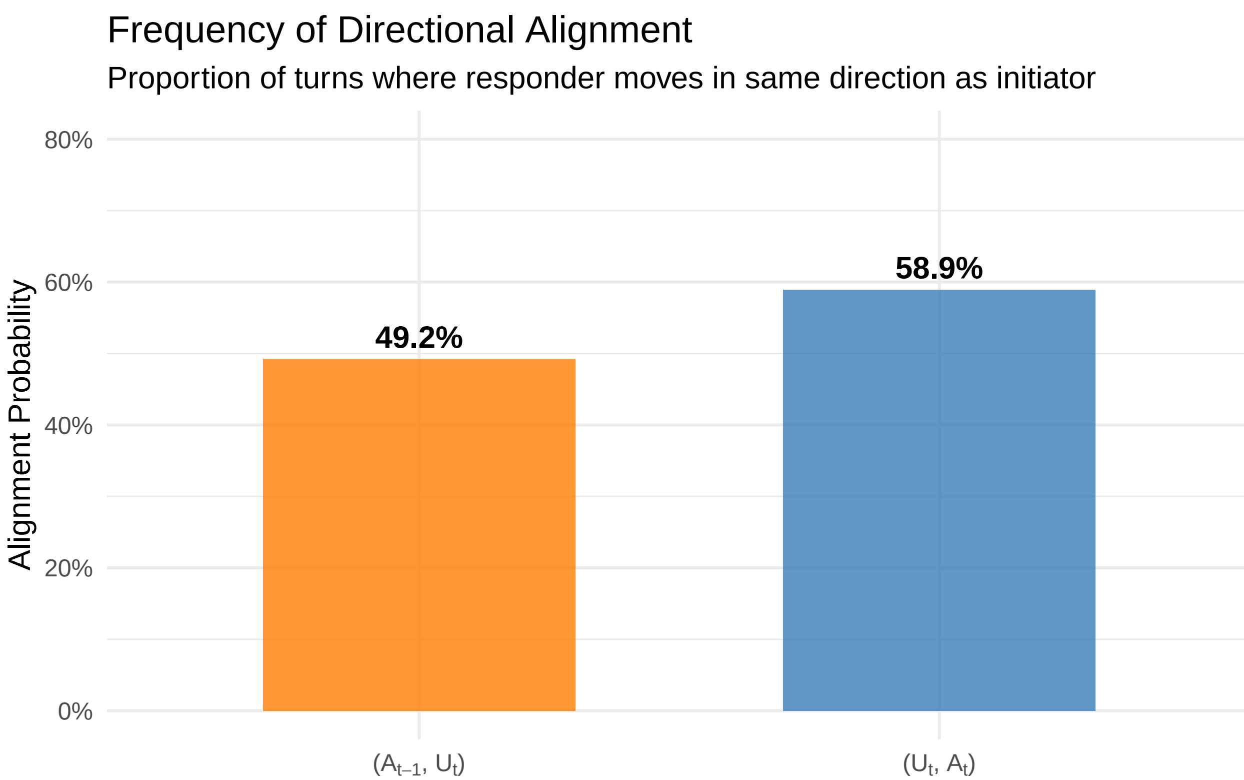
4.10 Valence Distance Difference Test

```
##
## === Valence Distance Model: User (Self vs Interlocutor) ===

##                        Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)              0.3802     0.0145  94.256 26.2178   0.0000
## directionInterlocutor   -0.0126     0.0155 686.747 -0.8129   0.4166

##
## === Valence Distance Model: AI (Self vs Interlocutor) ===

##                        Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)              0.3478     0.0106 1151.3136 32.7421   0.0000
## directionInterlocutor   -0.0056     0.0166   61.3057 -0.3381   0.7365
```



4.11 4.7 Alignment Duration and Stability

4.11.1 4.7.1 Visualization: Alignment Frequency (Stability)

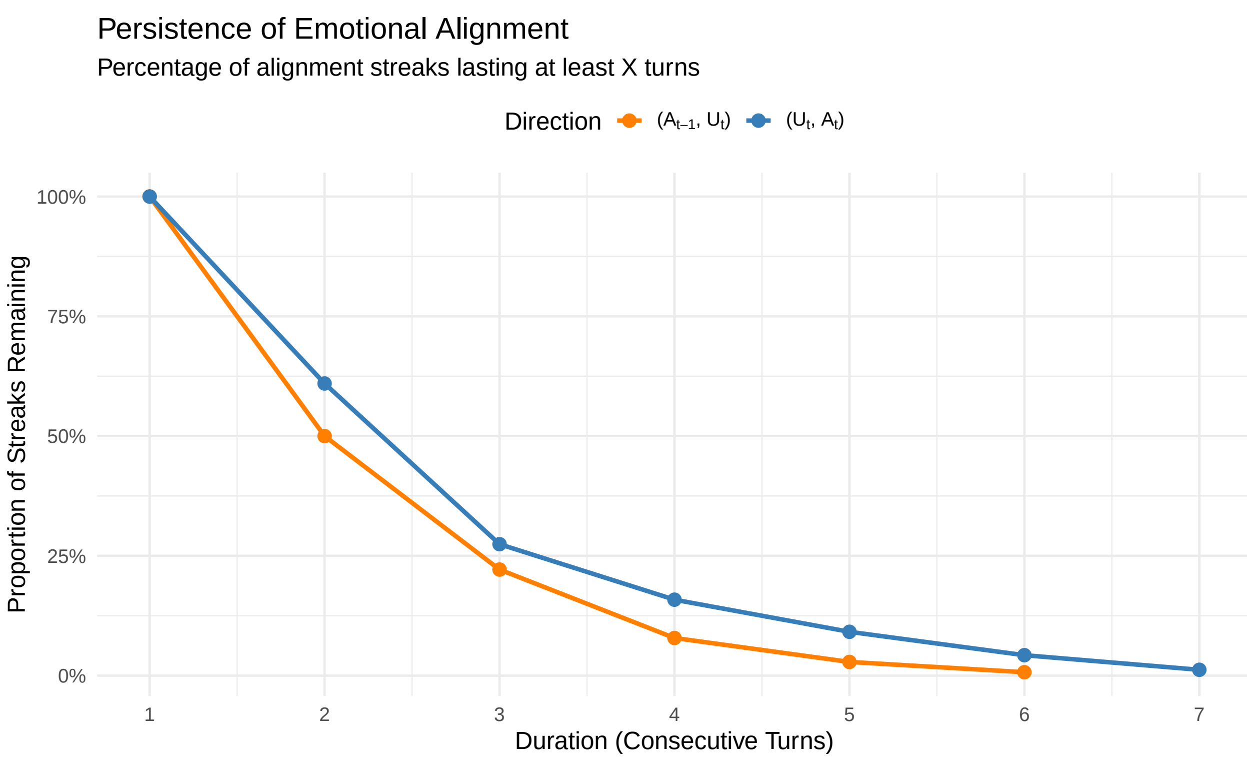



```
##
## === Proportions Test: Alignment Frequency ===
##
##  2-sample test for equality of proportions with continuity correction
##
```

```
## data:  alignment_frequency$aligned_turns out of alignment_frequency$total_turns
## X-squared = 10.309, df = 1, p-value = 0.001324
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  -0.15694805 -0.03735956
## sample estimates:
##    prop 1    prop 2
## 0.4923372 0.5894910
```

4.11.2 4.7.2 Visualization: Duration Distribution (Survival Analysis)

```
## # A tibble: 2 x 5
##   direction_label mean_duration median_duration max_duration n_streaks
##   <chr>                   <dbl>           <dbl>        <int>     <int>
## 1 prev_turn                1.84             1.5            6       140
## 2 same_turn                2.19             2              7       164
```




```
##
## === Wilcoxon Rank Sum Test: Alignment Duration ===
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  duration by direction_label
## W = 10002, p-value = 0.03943
## alternative hypothesis: true location shift is not equal to 0
```

5 6. Summary and Conclusions

5.1 6.1 Key Findings Summary

```
## # A tibble: 5 x 3
##   Metric                          User    LLM
##   <chr>                          <dbl>  <dbl>
## 1 Mean Valence                  -0.089  0.004
## 2 Sentiment Alignment ( )        0.232  0.091
## 3 Alignment Frequency (%)       49.2   58.9
## 4 Mean Alignment Duration (turns)  1.84   2.19
## 5 Semantic Alignment (Cosine)    0.678  0.691
```

6 7. Supplementary: Interaction Effects

6.1 7.1 Starter Interaction Effects

6.1.1 7.1.1 Same-Turn Alignment by Starter (Removed for now)

```
##             Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)  -0.0389     0.0334  249.4249 -1.1633   0.2458
## turn         -0.0007     0.0040 1304.0000 -0.1724   0.8632
```

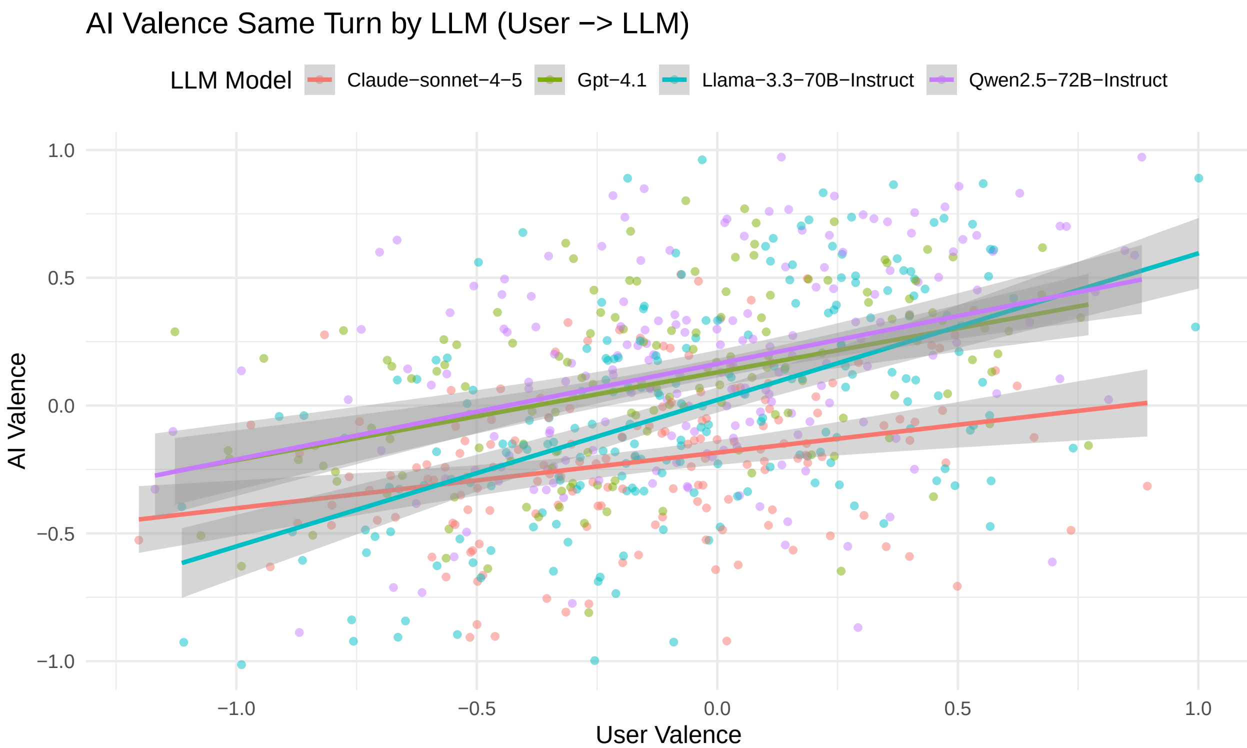
6.1.2 7.1.2 Previous-Turn Alignment by Starter (Removed for now)

6.1.3 7.1.3 Per-Story Correlations by Starter (Removed for now)

6.2 7.2 LLM Type Interaction Effects

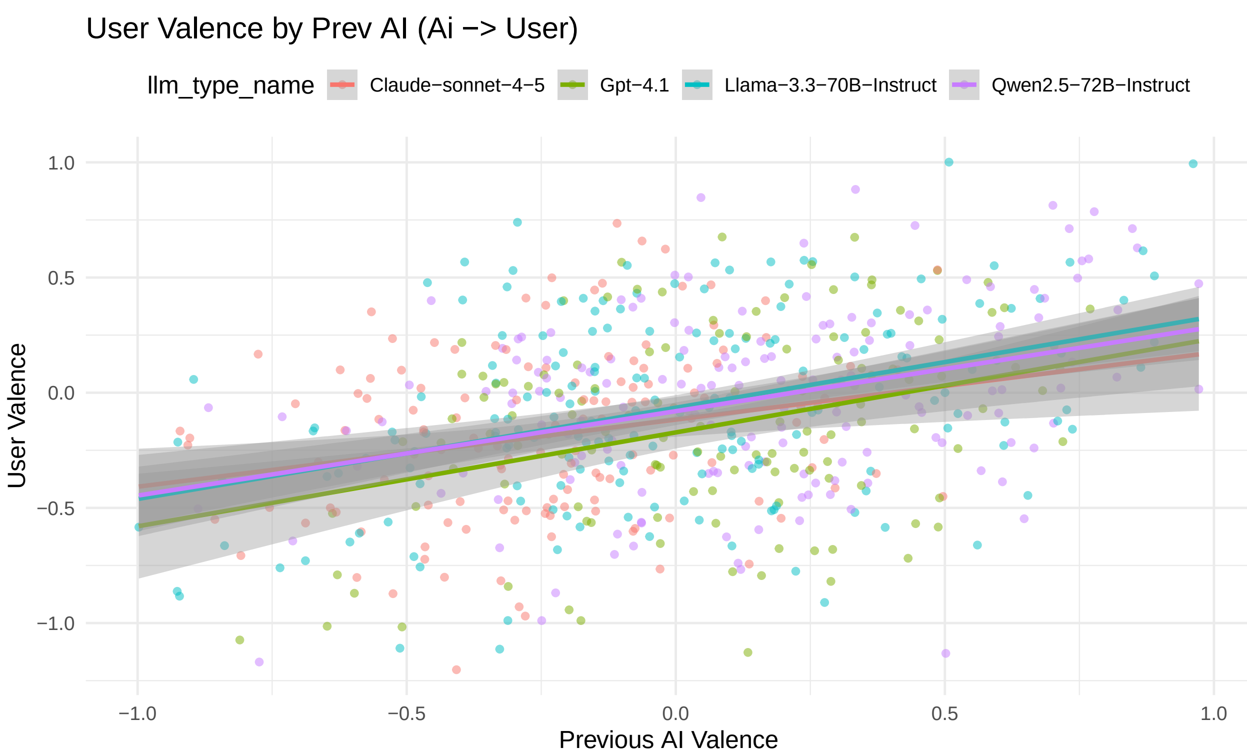
6.2.1 7.2.1 Same-Turn Alignment by LLM Type

```
## === Same-Turn Alignment: LLM Type Interaction ===

##                               Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                     0.1208     0.0449  80.1881  2.6891   0.0087
## user_valence                    0.2722     0.0682 676.4800  3.9916   0.0001
## factor(llm_type)2              -0.3146     0.0623  82.2939 -5.0474   0.0000
## factor(llm_type)3              -0.1071     0.0581  78.6317 -1.8434   0.0690
## factor(llm_type)4               0.0399     0.0595  77.9823  0.6709   0.5042
## user_valence:factor(llm_type)2 -0.1164     0.0975 686.8242 -1.1946   0.2327
## user_valence:factor(llm_type)3  0.1684     0.0919 666.1010  1.8313   0.0675
## user_valence:factor(llm_type)4 -0.0032     0.0932 685.0224 -0.0347   0.9723
```



6.2.2 7.2.2 Previous-Turn Alignment by LLM Type

```
## === Previous-Turn Alignment: LLM Type Interaction ===

##                                  Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                       -0.1634     0.0515  74.5484 -3.1731   0.0022
## ai_valence_lag1                    0.2768     0.1031 584.9247  2.6850   0.0075
## factor(llm_type)2                  0.0243     0.0743  88.1319  0.3267   0.7447
## factor(llm_type)3                  0.0962     0.0667  73.6060  1.4412   0.1538
## factor(llm_type)4                  0.0965     0.0694  76.7147  1.3911   0.1682
## ai_valence_lag1:factor(llm_type)2 -0.0849     0.1457 600.0333 -0.5826   0.5604
## ai_valence_lag1:factor(llm_type)3 -0.0596     0.1246 581.6830 -0.4787   0.6323
## ai_valence_lag1:factor(llm_type)4 -0.0047     0.1298 583.9839 -0.0362   0.9711
```



6.2.3 7.2.3 Per-Story Correlations by LLM Type

```
## === Same-Turn Correlations by LLM Type ===

## # A tibble: 4 x 4
##   llm_type mean_rho sd_rho n_stories
##      <int>    <dbl>  <dbl>     <int>
## 1        1    0.301  0.353        18
## 2        2    0.198  0.364        20
## 3        3    0.255  0.405        26
## 4        4    0.180  0.432        23

##
## ANOVA (LLM type effect on same-turn rho):

##                   Df Sum Sq Mean Sq F value Pr(>F)
## factor(llm_type)   3  0.185 0.06177   0.399  0.754
## Residuals         83 12.846 0.15477
```

6.3 7.3 Valence Distribution by Conditions

6.3.1 7.3.1 Valence by Starter

6.3.2 7.3.2 Valence by LLM Type

```
## # A tibble: 8 x 5
##   llm_type agent        mean_valence sd_valence n_turns
##      <int> <chr>               <dbl>      <dbl>   <int>
## 1        1 ai_valence         0.0854      0.344     144
## 2        1 user_valence      -0.130       0.428     144
## 3        2 ai_valence        -0.219       0.296     160
## 4        2 user_valence      -0.159       0.381     160
## 5        3 ai_valence        -0.0146      0.425     208
```

```
## 6        3 user_valence      -0.0641      0.404     208
## 7        4 ai_valence         0.154       0.394     184
## 8        4 user_valence      -0.0246      0.393     184
```

6.4 8. Session Information

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.3
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK ve
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Australia/Melbourne
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_2.0.1   here_1.0.2      afex_1.4-1      cowplot_1.1.3
##  [5] patchwork_1.3.0 zoo_1.8-12      stringi_1.8.4   arrow_19.0.1.1
##  [9] lmerTest_3.1-3  lme4_1.1-35.5   Matrix_1.7-1    lubridate_1.9.3
## [13] forcats_1.0.0   stringr_1.5.1   dplyr_1.1.4     purrr_1.0.2
## [17] readr_2.1.5     tidyr_1.3.1     tibble_3.2.1    ggplot2_4.0.2
## [21] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    farver_2.1.2        S7_0.2.1
##  [4] fastmap_1.2.0       TH.data_1.1-2       pacman_0.5.1
##  [7] digest_0.6.37       timechange_0.3.0    estimability_1.5.1
## [10] lifecycle_1.0.5     survival_3.7-0      magrittr_2.0.3
## [13] compiler_4.4.2      rlang_1.1.7         tools_4.4.2
## [16] utf8_1.2.4          yaml_2.3.10         knitr_1.50
## [19] labeling_0.4.3      bit_4.0.5           plyr_1.8.9
## [22] RColorBrewer_1.1-3  abind_1.4-8         multcomp_1.4-26
## [25] withr_3.0.2         numDeriv_2016.8-1.1 grid_4.4.2
## [28] fansi_1.0.6         xtable_1.8-4        scales_1.4.0
## [31] MASS_7.3-61         tinytex_0.52        cli_3.6.5
## [34] mvtnorm_1.3-3       rmarkdown_2.28      crayon_1.5.3
## [37] ragg_1.3.2          generics_0.1.3      rstudioapi_0.16.0
## [40] reshape2_1.4.4      tzdb_0.4.0          minqa_1.2.8
## [43] splines_4.4.2       assertthat_0.2.1    parallel_4.4.2
## [46] vctrs_0.7.1         boot_1.3-31         sandwich_3.1-1
## [49] carData_3.0-5       car_3.1-3           hms_1.1.3
## [52] pbkrtest_0.5.3      bit64_4.0.5         Formula_1.2-5
## [55] systemfonts_1.1.0   glue_1.8.0          nloptr_2.1.1
## [58] codetools_0.2-20    gtable_0.3.6        pillar_1.9.0
## [61] htmltools_0.5.8.1   R6_2.6.1            textshaping_0.4.0
## [64] rprojroot_2.1.1     vroom_1.6.5         evaluate_1.0.3
```

```
## [67] lattice_0.22-6      backports_1.5.0     broom_1.0.6
## [70] Rcpp_1.0.13         coda_0.19-4.1       nlme_3.1-166
## [73] mgcv_1.9-1          xfun_0.54           pkgconfig_2.0.3
```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.